

NMP2L2812
June 03, 2022

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Nine Mile Point Nuclear Station, Unit 2
Renewed Facility Operating License No. NPF-69
Docket No. 50-410

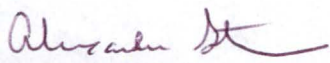
Subject: NMP2 Licensee Event Report 2022-001, Revision 0, Automatic Reactor
Scram due to Low Reactor Water Level During Maintenance

In accordance with the reporting requirements contained in 10 CFR 50.73(a)(2)(v)(B), please find enclosed NMP2 Licensee Event Report (LER) 2022-001, Revision 0, Automatic Reactor Scram due to Low Reactor Water Level During Maintenance

There are no regulatory commitments contained in this letter.

Should you have any questions regarding the information in this submittal, please contact Brandon Shultz, Site Regulatory Assurance Manager, at (315) 349-7012.

Respectfully,


Alexander Sterio
Plant Manager, Nine Mile Point Nuclear Station

AS/JA

Enclosure: NMP2 Licensee Event Report 2022-001, Revision 0, Automatic Reactor
Scram due to Low Reactor Water Level During Maintenance

cc: NRC Regional Administrator, Region I
NRC Resident Inspector
NRC Project Manager

IE22
NRR

Enclosure

NMP2 Licensee Event 2022-001, Revision 0
Automatic Reactor Scram due to Low Reactor Water Level During Maintenance
Nine Mile Point Nuclear Station, Unit 2

Renewed Facility Operating License No. NPF-69



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk all: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name	2. Docket Number	3. Page
Nine Mile Point Unit 2	05000 410	1 OF 5

4. Title
Automatic Reactor Scram due to Low Reactor Water Level During Maintenance

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
04	05	2022	2022	- 001 -	00	06	03	2022	N/A	05000 N/A
									Facility Name	Docket Number
									N/A	05000 N/A

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	
☐ OTHER (Specify here, in abstract, or NRC 366A).				

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Brandon Shultz, Site Regulatory Assurance Manager	(315) 349-7012

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
D	N/A	N/A	N/A	Y	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected		15. Expected Submission Date		Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)			NA	NA	NA

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On Thursday, April 5, 2022, at approximately 0222 hours with power level at approximately 100 percent, Nine Mile Point Unit 2 (NMP2) experienced a feedwater level control transient during maintenance on the Feedwater Flow Control Valve (LV10B), which resulted in a low reactor water level scram. This event is reportable per 10CFR 50.73(a)(2)(iv)(A) as any event or condition that resulted in a manual or automatic actuation of any of the systems listed in 10CFR 50.73(a)(2)(iv)(B). Following the reactor scram, all plant systems responded per design.

The root cause of this event was incorrect procedure guidance for nulling the demand and position signal in N2-SOP-06, "Feedwater Failures, Attachment 1, Restoration of 2FWS-LV10 Control". Corrective actions included revising N2-SOP-06, Attachment 1 to correctly null the demand signal to the potentiometer intended for control.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Nine Mile Point Unit 2	05000- 410	YEAR	SEQUENTIAL NUMBER	REV NO.
		2022	001	00

NARRATIVE**I. DESCRIPTION OF EVENT****A. PRE-EVENT PLANT CONDITIONS:**

Prior to the event, NMP2 was in Mode 1, operating at approximately 100 percent rated thermal power. Feedwater level control was controlling in automatic with Feedwater Level Control Valve LV10A controlling in AUTO on the backup position sensor due to the primary position sensor being non-functional. Feedwater Level Control Valve LV10B was controlling in MANUAL with both primary and backup position sensors non-functional.

Feedwater flow control is provided by a flow control valve in the discharge line of each reactor feed pump. The feedwater flow control valves may be controlled either in automatic or manual. Each feedwater flow control valve is provided with a primary and a secondary sensor that provides position feedback to the feedwater level control system.

B. EVENT:

On Thursday, April 5, 2022, at approximately 0222 hours, NMP2 experienced a reactor scram due to reactor water low level while performing maintenance on Feedwater Level Control Valve LV10B.

The potentiometer of the primary position sensor was being replaced on Feedwater Level Control Valve LV10B. Feedwater Level Control Valve LV10B was placed in a lock-up condition to support replacement of the potentiometer primary position sensor to allow Operations to regain automatic valve functionality. After the potentiometer was installed, Operations attempted to restore Feedwater Level Control Valve LV10B to the primary position using N2-SOP-06, Feedwater Failures, Attachment 1. Upon restoring Feedwater Level Control Valve LV10B primary positioner, the position feedback signal was mismatched (80% open and demanded position was 43%), causing Feedwater Level Control Valve LV10B to rapidly close which subsequently caused reactor water level to lower. Feedwater Level Control Valve LV10A attempted to open to maintain reactor water level but was unsuccessful, resulting in a low reactor water level scram. High Pressure Core Spray and Reactor Core Isolation Cooling systems actuated as designed to maintain level, and plant personnel successfully recovered feedwater level control.

The condition was reported to the NRC on April 5, 2022 at approximately 0608 hours pursuant to the requirements of 10CFR50.72(b)(2)(iv)(A), 50.72(b)(2)(iv)(B), and 50.72(b)(3)(iv)(A) (Event Notification #55821)

Nine Mile Point Unit 1 (NMP1) was unaffected.

The event has been entered into the plant's corrective action program as IR 04490223.

C. INOPERABLE STRUCTURES, COMPONENTS, OR SYSTEMS THAT CONTRIBUTED TO THE EVENT:

No other systems, structures, or components contributed to this event.

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Nine Mile Point Unit 2	05000-410	2022	001	00

NARRATIVE**D. DATES AND APPROXIMATE TIMES OF MAJOR OCCURRENCES AND OPERATOR ACTIONS:**

The dates, times, major occurrences, and operator actions for this event are as follows.

April 2, 2022:

2249 - The primary position feedback for Feedwater Level Control Valve LV10B failed

April 5, 2022:

0000 - Shortly after midnight, Instrument Maintenance Department (IMD) began the prejob brief and maintenance activity for replacement of the Feedwater Level Control Valve LV10B primary potentiometer

0137 - Operations unlocked Feedwater Level Control Valve LV10B by taking the S10B switch to NORMAL to allow IMD technicians to get initial position readings. Once readings were captured and documented, Operations placed the valve back into the lock-up condition by moving the S10B switch to reset per N2-SOP-06. IMD then matched the replacement potentiometer resistance to the originally installed potentiometer and installed the new primary potentiometer.

0222 - After completion of the primary potentiometer replacement and adjustment, Operations repositioned the control switch to NORMAL, resetting the lockup condition of the valve and returning Feedwater Level Control Valve LV10B to service. Upon returning to service, the feedwater flow control system erroneously read the Feedwater Level Control Valve LV10B position at approximately 80% versus the required demand signal of 43% causing it to drive the valve closed. Feedwater Level Control Valve LV10A responded however, reactor water level lowered and scrambled on Low Water Reactor Water Level.

0224 - Turbine tripped

E. METHOD OF DISCOVERY:

This event was discovered by Reactor Operators when feedwater flow control valves began moving and Reactor scram annunciation was received.

F. SAFETY SYSTEM RESPONSES:

All safety systems responded per design.

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NARRATIVE**II. CAUSE OF THE EVENT**

The root cause of this event was determined to be incorrect procedure guidance for nulling the demand and position signal in procedure N2-SOP-06, Feedwater Failures, Attachment 1, Restoration of 2FWS-LV10 Control. When the NORMAL/RESET switch (S10B) is placed to reset, the signal sent to the M/A (manual-automatic) station is from the non-selected position feedback potentiometer and not the potentiometer selected with the corresponding Feedback Potentiometer selector switch (S2B). This procedure error was introduced when the modification was implemented and not recognized.

III. ANALYSIS OF THE EVENT

The automatic reactor scram is reportable per 10CFR 50.72(b)(2)(iv)(B) and 10CFR 50.73(a)(2)(iv)(A). It is defined under paragraph 10CFR 50.73(a)(2)(iv)(A) as any event or condition that resulted in a manual or automatic actuation of any of the systems listed in 10CFR 50.73(a)(2)(iv)(B).

Loss of feedwater transient events are considered in the design basis of the plant with multiple automatic and manual recovery paths. The High Pressure Core Spray and Reactor Core Isolation Cooling systems actuated as designed to maintain level, and plant personnel successfully recovered feedwater control shortly after the trip. It is judged that the safety significance of this event is low, and the event did not pose a threat to the health and safety of the public or plant personnel. All other plant systems performed per design without complications.

IV. CORRECTIVE ACTIONS**A. ACTION TAKEN TO RETURN AFFECTED SYSTEMS TO PRE-EVENT NORMAL STATUS:**

A new model of potentiometers were installed on Feedwater Level Control Valve LV10A and LV10B.

B. ACTION TAKEN OR PLANNED TO PREVENT OCCURRENCE:

Revised N2-SOP-06, Attachment 1 to correctly null the demand signal to the potentiometer intended for control.



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Nine Mile Point Unit 2	05000-410	YEAR 2022	SEQUENTIAL NUMBER 001	REV NO. 00

NARRATIVE

V. ADDITIONAL INFORMATION

A. FAILED COMPONENTS:

No component failures were associated with this event. An IRIS entry was completed for the event. (#525435)

B. PREVIOUS LERs ON SIMILAR EVENTS:

None.

C. THE ENERGY INDUSTRY IDENTIFICATION SYSTEMS (EIIS) COMPONENT FUNCTION IDENTIFIER AND SYSTEM NAME OF EACH COMPONENT OR SYSTEM REFERRED TO IN THIS LER:

COMPONENT	IEEE 803 FUNCTION IDENTIFIER	IEEE 805 SYSTEM IDENTIFICATION
Feedwater Level Control System	N/A	JB
Feedwater System	N/A	SJ
Reactor Protection System	N/A	JC
High Pressure Coolant Injection System	N/A	BJ
Reactor Core Isolation Cooling	N/A	BN

D. SPECIAL COMMENTS:

None